

RAG Evaluation Reference

Ground-truth questions for testing the Technical Document Q&A; Copilot

1. Evaluation Method

Evaluation should check whether the retriever finds the correct evidence and whether the generator answers faithfully from that evidence. The questions in this document are designed to test concepts contained in the companion technical references.

For retrieval evaluation, record the top three chunks returned for each question. For generation evaluation, compare the response against the expected answer and verify that no unsupported details were introduced.

2. Retrieval Test Categories

Category	What it tests
Definition	Whether a concept is explained correctly
Parameter	Whether exact configuration values are retrieved
Architecture	Whether pipeline relationships are retrieved
Implementation	Whether code-oriented details are found
Security	Whether operational guidance is grounded

3. Expected Grounding Behavior

When a question asks for information that is absent from the indexed documents, the correct behavior is to state that the provided documents do not contain enough information. The model should not fill the gap using general knowledge.

When an answer is present, the response should be concise and should ideally identify the source document and page number when the application exposes metadata.

4. Suggested Experiment

Run the same questions with top-k equal to 1, 3, and 5. Compare whether additional chunks improve answer coverage or introduce distracting information. This experiment demonstrates why retrieval depth is an important RAG design parameter.

A second experiment is to change chunk size while keeping the embedding model constant. Observe how chunk boundaries affect the evidence returned for implementation questions.

5. Acceptance Criteria

For this educational project, the initial system should retrieve the relevant technical passage for most in-scope questions, answer only from retrieved context, preserve source metadata, and explicitly refuse to invent an answer when evidence is unavailable.

6. Next Engineering Steps

After ingestion has been validated, create embeddings with all-MiniLM-L6-v2, persist the chunks in ChromaDB, implement top-3 semantic retrieval, connect an LLM with a grounded prompt, and expose the workflow through Streamlit.